

Problem Set 3

1

It turns out the toilet bowl acts as a resonant cavity at a nearby radio station. To understand the resonant electromagnetic modes of the smart toilet, you are asked to determine the volume of the bowl. Refer to the drawing below:

a

The bowl volume is controlled by three variables (R, W, L). Set up a volumetric integral to calculate the volume V for a given (R, W, L). NOTE: R refers to the radius of the semicircle denoted by that dimension.

✓ Answer ✓

We can calculate the area of the hemisphere and the hemicylinder separately

$$\iiint_S 1 \, dS = \int_0^{2\pi} \int_0^{\pi/2} \int_0^R \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$
$$\iiint_C 1 \, dC = \int_0^H \int_{-\pi/2}^{\pi/2} \int_0^{D/2} r \, dr \, d\theta \, dz$$

We can recognize the first as the volume of a hemisphere and the second as a volume of a hemicylinder

$$\frac{2}{3}\pi R^3 + \frac{1}{8}\pi D^2 H$$

b

The US EPA wants to know how much water the toilet bowl can hold. What is the maximum volume of the bowl using the nominal dimensions in the drawing?

✓ Answer

$$84823 \text{ cm}^3$$

2.23

A load with impedance $Z_L = (25 - j50)\Omega$ is to be connected to a lossless transmission line with the characteristic impedance Z_0 . Z_0 is chosen such that the standing-wave ratio is the smallest possible. What should Z_0 be?

✓ Answer

To minimize standing waves, we should use the magnitude of our load.

$$|Z_L| = \sqrt{25^2 + 50^2} \approx 55.9 \Omega$$

2.28

A lossless transmission line of electrical length $l = 0.35\lambda$ is terminated in a load impedance as shown in Fig. P2.28. Find Γ , S , and Z_{in} . Verify your results using Modules 2.4 or 2.5. Include a printout of the screen's output display.

$$Z_L = 60 + j30 \Omega$$

$$Z_0 = 100 \Omega$$

✓ **Answer**

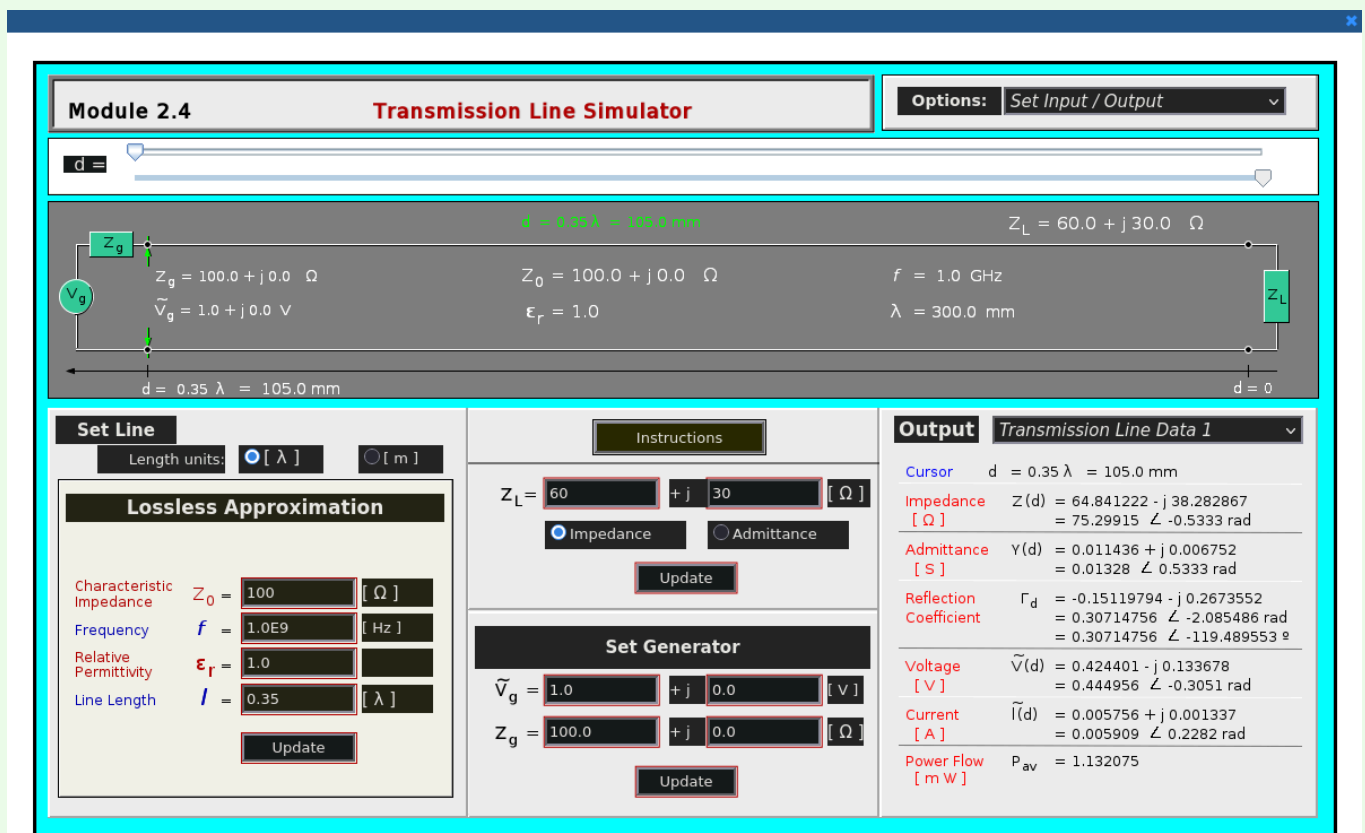
$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = -0.20754716981132076 + 0.22641509433962265j$$

$$|\Gamma| = 0.3071475584169756$$

$$S = \frac{1+\Gamma}{1-\Gamma} = 0.6 + 0.3j$$

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tan \beta l}{Z_0 + Z_L \tan \beta l}$$

$$= 64.8412219022718 - 38.28286695032168j$$



2.40

A 100 MHz FM broadcast station uses a 300Ω transmission line between the transmitter and a tower-mounted half-wave dipole antenna. The antenna impedance is 73Ω . You are asked to design a quarter-wave transformer to match the antenna to the line.

a

Determine the electrical length and characteristic impedance of the quarter-wave section.

✓ **Answer**

$$l = \lambda/4 = 90^\circ$$

$$Z_C = \sqrt{Z_0 Z_L} = 147.99 \, \Omega$$

b

If the quarter-wave section is a two-wire line with $D = 2.5 \, \text{cm}$ and the wires are embedded in polystyrene with $\epsilon_r = 2.6$, determine the physical length of the quarter-wave section and the radius of the two wire conductors.

✓ **Answer**

$$\lambda = \frac{u_p}{f} = \frac{c}{\sqrt{\epsilon_r} f} = 1.8605210188381267 \, \text{m}$$

$$\lambda/4 = 0.4651302547095317 \, \text{m}$$

$$Z_0 = 148 = \frac{120}{\sqrt{\epsilon_r}} \cosh^{-1}(D/d)$$

$$r = 0.335893601488105 \, \text{cm}$$

3.15

A certain plane is described by

$$2x + 3y + 4z = 16$$

Find the unit vector normal to the surface in the direction away from the origin.

✓ **Answer**

$$n = \langle 2, 3, 4 \rangle$$

$$|n| = \sqrt{29}$$

$$n = \langle 2/\sqrt{29}, 3/\sqrt{29}, 4/\sqrt{29} \rangle$$

3.50

A vector field $D = \hat{r}r^3$ exists in the region between two concentric cylindrical surfaces defined by $r = 1$ and $r = 2$ with both cylinders extending between $z = 0$ and $z = 5$. Verify the divergence theorem by evaluating the following:

a

$$\oint_S D \cdot ds$$

✓ Answer

Outer surface:

$$\int_0^5 \int_0^{2\pi} 16 \, d\phi \, dz = 160\pi$$

Inner surface:

$$\int_0^5 \int_0^{2\pi} -1 \, d\phi \, dz = -10\pi$$

Top and bottom surfaces:

$$D \cdot ds = 0 \text{ (perpendicular)}$$

Total:

$$160\pi - 10\pi = 150\pi$$

b

$$\int_V \nabla \cdot D \, dv$$

✓ Answer

$$\nabla \cdot D = 4r^2$$

$$\int_0^5 \int_0^{2\pi} \int_1^2 4r^3 \, dr \, d\theta \, dz = 150\pi$$

They are in fact the same